Self-Calibrating Monocular Geometry at Test Time

Estimating 3D geometry from a single image is fundamentally ambiguous, and the quality of the reconstruction is closely related to camera parameters such as focal length and field of view. MoGe is a recent monocular geometry model that predicts dense 3D point maps, depth, and camera information directly from a single open-domain image.

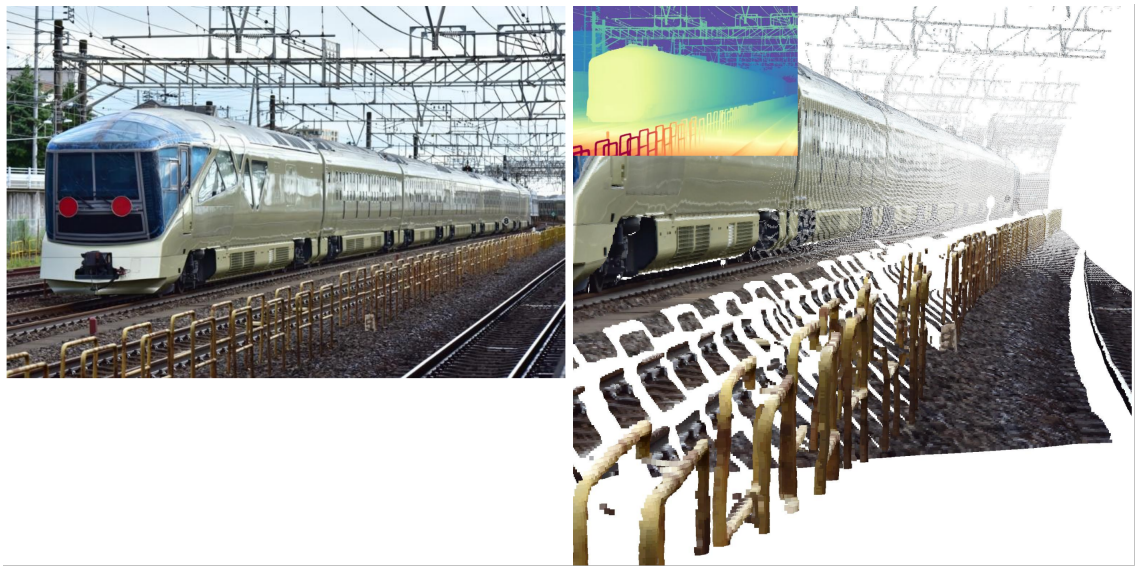


Figure 1: An example of monocular geometry estimation using MoGe.

In this project, you will investigate whether a pretrained monocular geometry model can improve its own camera estimation at test time without any additional training. Given one image, you may generate several geometrically related views through cropping, rescaling, or changes of image resolution. Since the geometric relationship between these transformed images is known, their predicted camera parameters should ideally be mutually consistent.

Your task is to design a training-free self-calibration strategy that combines the camera or field-of-view predictions from multiple transformed views. For example, you may develop a robust consensus rule, confidence-weighted fusion, or a consistency-based optimization method to obtain a refined camera estimate. The calibrated camera parameters can then be provided back to MoGe to obtain an improved depth or 3D point-map prediction.

You should evaluate whether self-calibration improves camera estimation and 3D geometry on public RGB-D datasets. Possible extensions include uncertainty-aware fusion, adaptive selection of useful crops, joint refinement of camera and geometry, or extending the same strategy to other monocular geometry models.

References

[1] R. Wang et al., "MoGe: Unlocking Accurate Monocular Geometry Estimation for Open-Domain Images with Optimal Training Supervision," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[2] Official MoGe implementation: https://github.com/microsoft/MoGe

[3] NYU Depth V2 Dataset: https://cs.nyu.edu/~fergus/datasets/nyu_depth_v2.html